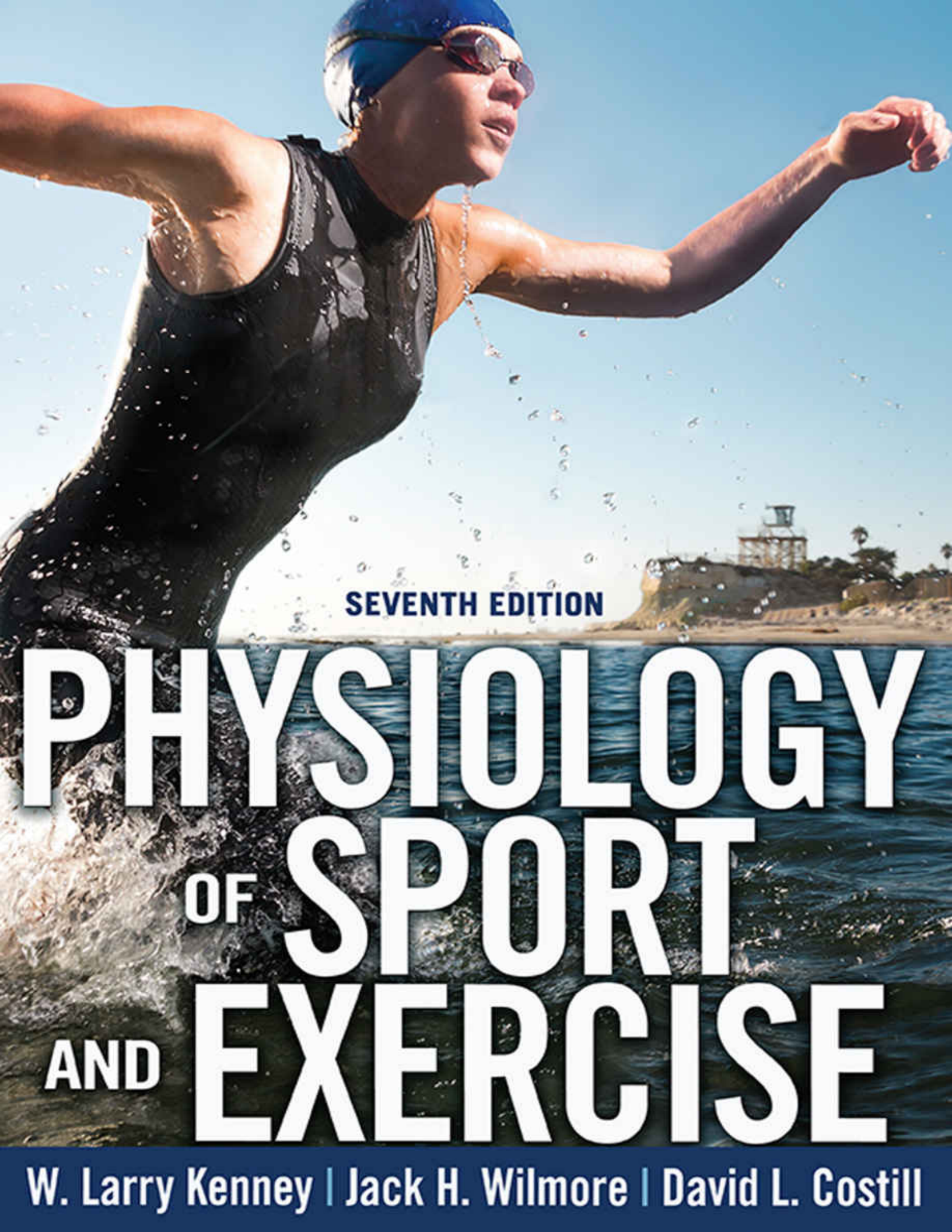




SEVENTH EDITION

PHYSIOLOGY OF SPORT AND EXERCISE

W. Larry Kenney | Jack H. Wilmore | David L. Costill

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PHYSIOLOGY OF SPORT AND EXERCISE

SEVENTH EDITION

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HUMAN KINETICS

Library of Congress Cataloging-in-Publication Data

Names: Kenney, W. Larry, author. | Wilmore, Jack H., 1938-2014, author. | Costill, David L., author.

Title: Physiology of sport and exercise / W. Larry Kenney, Jack H. Wilmore, David L. Costill.

Description: Seventh edition. | Champaign, IL : Human Kinetics, [2020] | Includes bibliographical references and index.

Identifiers: LCCN 2018040753 (print) | LCCN 2018041421 (ebook) | ISBN 9781492574859 (epub) | ISBN 9781492589198 (PDF) | ISBN 9781492572299 (hardback)

Subjects: | MESH: Sports--physiology | Exercise--physiology | Physical Fitness--physiology | Physical Endurance--physiology

Classification: LCC QP301 (ebook) | LCC QP301 (print) | NLM QT 260 | DDC 612/.044--dc23

LC record available at <https://lcn.loc.gov/2018040753>

ISBN: 978-1-4925-7229-9 (hardback)

ISBN: 978-1-4925-7486-6 (loose-leaf)

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The web addresses cited in this text were current as of January 2019, unless otherwise noted.

Senior Acquisitions Editor: Amy N. Tocco; **Developmental Editor:** Judy Park; **Managing Editor:** Anna Lan Seaman; **Copyeditor:** Joy Hoppenot; **Indexer:** Alisha Jeddelloh; **Permissions Manager:** Dalene Reeder; **Senior Graphic Designer:** Nancy Rasmus; **Cover Designer:** Keri Evans; **Cover Design**

Associate: Susan Rothermel Allen; **Photograph (cover):** AMR Image/Getty Images; **Photo Asset Manager:** Laura Fitch; **Visual Production Assistant:** Joyce Brumfield; **Photo Production Manager:** Jason Allen; **Senior Art Manager:** Kelly Hendren; **Illustrations:** © Human Kinetics, unless otherwise noted; **Printer:** Walsworth

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Printed in the United States of America 10 9 8 7 6 5 4 3 2 1

The paper in this book was manufactured using responsible forestry methods.

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P.O. Box 5076

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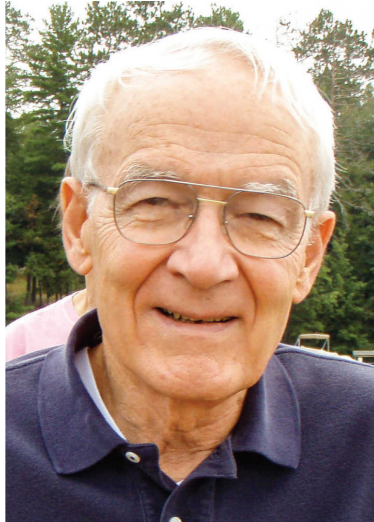
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Jack H. Wilmore

April 23, 1938 - November 15, 2014

Jack H. Wilmore was an exceptional teacher, researcher, writer, and lecturer. His ability to communicate the complexities of exercise physiology to students, health professionals, and the general public is evident in this textbook. As the lead author of the first four editions of *Physiology of Sport and Exercise*, Jack took great pride in the clarity and accuracy of its contents. This book was his brainchild.

Jack began his career in exercise physiology at Ithaca College in New York. He then held professorships at the University of California at Berkley, University of California at Davis, University of Arizona, University of Texas, and Texas A&M. He published more than 300 scientific and lay articles, 15 books, and 55 chapters in other texts. In addition to serving as president of the American College of Sports Medicine (ACSM) and the American Academy of Kinesiology and Physical Education, Jack was active in many other professional organizations. His star status in sports medicine was rewarded with a long list of honors including the Citation and Honor Awards from ACSM. The achievements in his 50-year career are the basis for our current knowledge of the critical importance of regular physical

activity in health, disease, and aging. His impact on students and the general public was the envy of all his colleagues. *Physiology of Sport and Exercise* lives on as a legacy of an exceptional scientist in sport and exercise and friend to many. He is missed by family, friends, and colleagues alike, and this book remains an enduring part of his legacy.

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Preface

Physiology is the study of how the human body functions. Cells, tissues, organs, and systems intricately and precisely communicate and integrate to coordinate the body's myriad physiological functions. Even at rest, the body is physiologically quite active. Imagine, then, how much more active all of these body systems become when you engage in exercise. During exercise, nerves excite muscles to contract. Exercising muscles are metabolically active and require more nutrients, more oxygen, and efficient clearance of waste products. The autonomic nervous system and endocrine glands combine to fine-tune these processes. How does the whole body respond to the increased physiological demands of physical activity in all its forms?

That is the key question when you study the physiology of sport and exercise. *Physiology of Sport and Exercise, Seventh Edition*, introduces you to the fields of sport and exercise physiology. Our goal is to build on the knowledge that you developed during basic coursework in human anatomy and physiology and to apply those principles in studying how the body (1) performs and responds to the added demands of an acute bout of exercise and (2) adapts to repeated bouts of exercise (i.e., exercise training).

What's New in the Seventh Edition

The seventh edition of *Physiology of Sport and Exercise* maintains the previous edition's high standard for illustrations, photos, and medical artwork. This visual detail, clarity, and realism allow both a greater insight into the physiological responses to exercise and a better understanding of the underlying research. In addition, the text is now augmented with animations, audio clips, and video clips, provided online in the student web study guide and separately for instructors. Throughout the text, you will find icons to identify pieces

of artwork that are the basis for an animation or that have an accompanying audio clip. Accessing these resources will further aid understanding of the illustrations and the physiological processes they represent. In addition, video clips feature experts in the field discussing exciting current topics of research.

The new edition also brings back the Research Perspective elements introduced in the last edition that highlight interesting current research. These inserts discuss a wide range of important new or developing topics in sport and exercise physiology, providing interested students with additional insight into the state of research in the field. We have also revised the introductory chapter to include information on new frontiers in exercise physiology in the 21st century such as genomics and epigenetics, [chapter 5](#) to include more detailed coverage of mechanisms associated with fatigue and muscle cramps, [chapter 11](#) to expand coverage of high-intensity interval training, and [chapter 17](#) to focus more on health benefits of physical activity in children and adolescents. In addition, we have extensively updated the text to include the latest research on important topics in the field, including the following:

- New information on length–tension and force–velocity relations in muscle ([chapter 1](#)) and individual variability in appetite hormones ([chapter 4](#))
- Newly added sections on the crossover concept ([chapter 2](#)), critical power ([chapter 5](#)), functional sympatholysis ([chapter 6](#)), the oxygen cascade ([chapters 7](#) and [13](#)), group exercise ([chapter 9](#)), and exercise and mobility in aging ([chapter 18](#))
- Updated information on the role of maximal stroke volume in determining maximal aerobic capacity ([chapter 8](#))
- Expanded mechanistic discussion of protein synthesis in muscle hypertrophy ([chapter 10](#))
- Revision of information to reflect published guidelines by professional organizations on nutrition and athletic performance ([chapter 15](#)), exercise and pregnancy ([chapter 19](#)), and health-related screening, fitness testing, and exercise prescription ([chapter 20](#))

All of these changes were made while retaining our emphasis on the ease of reading and understanding that have made this book the leading text for introducing students to this exciting field. The overall structure and progression of the text have been retained from the sixth edition. Our first focus is on muscle and how its needs are altered as an individual goes from a resting to an active state and how these needs are supported by—and interact with—other body systems. In later chapters we address principles of exercise training; considerations of environmental factors of heat, cold, and altitude; sport performance; and exercise for disease prevention.

Organization of the Seventh Edition

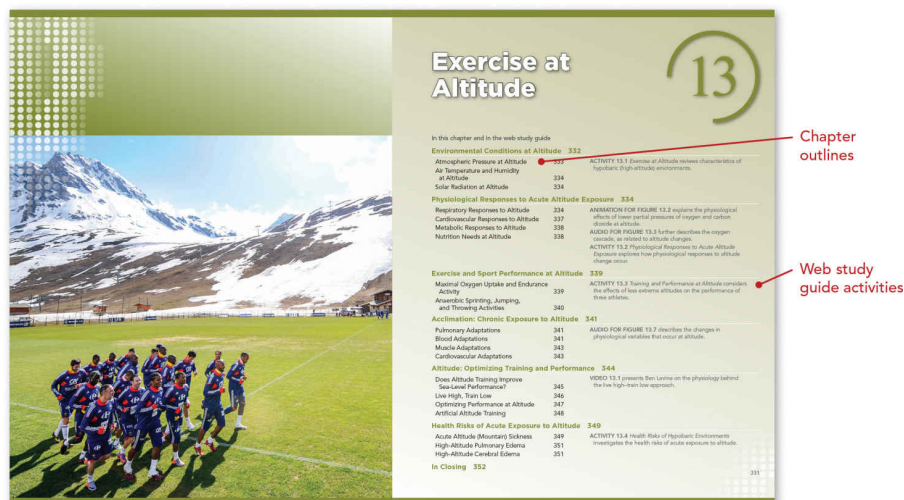
We begin in the introduction with a historical overview of sport and exercise physiology as they have emerged from the parent disciplines of anatomy and physiology, and we explain basic concepts that are used throughout the text. In [parts I](#) and [II](#), we review the major physiological systems, focusing on their responses to acute bouts of exercise. In [part I](#), we examine how the muscular, metabolic, nervous, and endocrine systems interact to produce body movement. In [part II](#), we look at how the cardiovascular and respiratory systems continue to deliver nutrients and oxygen to the active muscles and transport waste products away during physical activity. In [part III](#), we consider how these systems adapt to chronic exposure to exercise (i.e., training).

We change perspective in [part IV](#) to examine the impact of the external environment on physical performance. We consider the body's response to heat and cold, and then we examine the impact of low atmospheric pressure experienced at altitude. In [part V](#), we shift attention to how athletes can optimize physical performance. We evaluate the effects of different types and volumes of training. We consider the importance of appropriate body composition for optimal performance and examine athletes' special dietary needs, as well as how nutrition can be used to enhance performance. Finally, we explore the use of ergogenic aids—substances purported to improve athletic ability.

In [part VI](#), we examine unique considerations for specific populations. We look first at the processes of growth and

development and how they affect the performance capabilities of young athletes. We evaluate changes that occur in physical performance as people age and explore the ways in which physical activity can help maintain health and independence. Finally, we examine issues and special physiological concerns of female athletes.

In the final part of the book, [part VII](#), we turn our attention to the application of sport and exercise physiology to prevent and treat various diseases and the use of exercise for rehabilitation. We look at prescribing exercise for maintaining health and fitness, and we then close the book with a discussion of cardiovascular disease, obesity, and diabetes.



Special Features in the Seventh Edition

This seventh edition of *Physiology of Sport and Exercise* is designed with the goal of making learning easy and enjoyable. This text is comprehensive, but the many special features included will help you progress through the book without being overwhelmed by its scope. In addition to these features, the fully updated web study guide that accompanies this text provides opportunities for interactive learning and review, along with animations, audio clips, and video clips to enhance your understanding of the text.

Each chapter in the book begins with a chapter outline with page numbers to help you locate material. Also noted in the chapter

outline are the web study guide activities relating to each section of the chapter. Each chapter begins with a brief story that explores a real-world application of the concepts presented.

Within each chapter, the Research Perspective elements introduce you to important topics in current exercise physiology research. You will also find icons to alert you to animations and audio clips that will help you understand important figures and to video clips that provide expanded discussion on current topics in the field:

 **Animation** icons identify figures that are also provided as animations.

 **Audio** icons identify figures that are further explained in an accompanying audio clip.

 **Video** icons let you know when a video clip on a topic is available.

As you read through, you will also find In Review elements that summarize the major points presented in the previous sections. And at the end of the chapter, the In Closing wraps things up and notes how what you have learned sets the stage for the topics to come.

RESEARCH PERSPECTIVE 7.1

Sprint Interval Training for Respiratory Muscles

Typical respiratory muscle endurance training (RMET) improves exercise capacity and thus performance; such improvements are largely attributed to reductions in the development of respiratory muscle fatigue. However, can a shorter version of RMET based on the principle of high-intensity interval training (respiratory muscle sprint-interval training, or RMSIT) elicit similar improvements in respiratory muscle function?

A team of investigators recently sought to compare the effects of traditional RMET versus RMSIT on respiratory muscle function.³ Mechanical airway properties and respiratory muscle testing (e.g., respiratory muscle strength) were measured before and after experimental sessions of RMET and RMSIT. RMET consisted of continuous volitional hyperpnea (increased depth and rate of breathing) performed for 30 min using a commercially available training device. The RMSIT was a novel respiratory muscle training regimen developed by the researchers. Using the same training device as with RMET, the RMSIT regimen consisted of six short maximal respiratory sprints with additional airway resistance to maximize respiratory muscle work. In this fashion, RMSIT is characterized by higher respiratory muscle power output and tension-time indices, but considerably lower total work compared to RMET. The standard RMET and the novel RMSIT regimens reduced respiratory muscle contractility to the same extent, triggering similar muscular adaptations in response to training. Neither protocol altered mechanical airway properties. Therefore, RMSIT appears to be a safe and time-saving alternative to RMET.

RMET can improve overall function for individuals who have undergone a median sternotomy (splitting of the sternum to access underlying organs) during cardiac surgery. Clinical exercise physiologists are interested in exercise training adaptations that occur with structured cardiac rehabilitation programs. The results of a 2013 study suggest that it would be beneficial to include exercises that improve the strength of the inspiratory muscles as part of a cardiac rehabilitation program.⁴ This type of training would reduce inspiratory muscle effort and further improve ventilatory efficiency in patients after open-heart surgery.

Research Perspective

alveoli into the blood in the pulmonary capillaries, and carbon dioxide diffuses from the blood into the alveoli in the lungs. The **alveoli** are grape-like clusters, or air sacs, at the ends of the terminal bronchioles.

Blood from the body (except for that returning from the lungs) returns through the vena cava to the right side of the heart. From the right ventricle, this blood is pumped through the pulmonary artery to the lungs, ultimately working its way into the pulmonary capillaries. These capillaries form a dense network around the alveolar sacs and are so small that the red blood cells must pass through them in single file, such that the maximal surface area of each cell is exposed to the surrounding lung tissue. This is where pulmonary diffusion occurs.

Blood Flow to the Lungs at Rest

At rest the lungs receive approximately 4 to 6 L/min of blood flow, depending on body size. Because cardiac output from the right side of the heart approximates cardiac output from the left side of the heart, blood flow to the lungs matches blood flow to the systemic circulation. However, pressure and vascular resistance in the blood vessels in the lungs are different than in the systemic circulation. The mean pressure in the pulmonary artery is ~15 mmHg (systolic pressure is ~25 mmHg and diastolic pressure is ~8 mmHg) compared to the mean pressure in the aorta of ~95 mmHg. The

pressure in the left atrium, where blood is returning to the heart from the lungs, is ~5 mmHg; thus, there is not a great pressure difference across the pulmonary circulation (15 – 5 mmHg). Figure 7.4 illustrates the differences in pressures between the pulmonary and systemic circulation.

Recalling the discussion of blood flow in the cardiovascular system from chapter 6, pressure = flow $\times$ resistance. Since blood flow to the lungs is equal to that of the systemic circulation, and there is a substantially lower change in pressure across the pulmonary vascular system, resistance is proportionally lower compared to that in the systemic circulation. This is reflected in differences in the anatomy of the vessels in the pulmonary versus systemic circulation: The pulmonary blood vessels are thin walled, with relatively little smooth muscle.

Respiratory Membrane

Gas exchange between the air in the alveoli and the blood in the pulmonary capillaries occurs across the **respiratory membrane** (also called the alveolar-capillary membrane). This membrane, depicted in figure 7.5, is composed of

- the alveolar wall,
- the capillary wall, and
- their respective basement membranes.

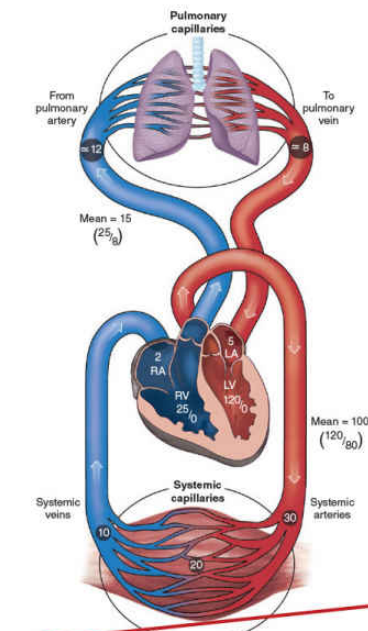


FIGURE 7.4 Comparison of pressures (mmHg) in the pulmonary and systemic circulations.

The primary function of these membranous surfaces is for gas exchange. The respiratory membrane is very thin, measuring only 0.5 to 4 μm . As a result, the gases in the nearly 300 million alveoli are in close proximity to the blood circulating through the capillaries.

Partial Pressures of Gases

The air we breathe is a mixture of gases. Each exerts a pressure in proportion to its concentration in the gas mixture. The individual pressures from each gas in a mixture are referred to as **partial pressures**. According to **Dalton's law**, the total pressure of a mixture of gases

equals the sum of the partial pressures of the individual gases in that mixture.

Consider the air we breathe. It is composed of 79.04% nitrogen (N_2), 20.93% oxygen (O_2), and 0.03% carbon dioxide (CO_2). These percentages remain constant regardless of altitude. At sea level, the atmospheric (or barometric) pressure is approximately 760 mmHg, which is also referred to as standard atmospheric pressure. Thus, if the total atmospheric pressure is 760 mmHg, then the partial pressure of nitrogen (P_{N_2}) in air is 600.7 mmHg (79.04% of the total 760 mmHg pressure). Oxygen's partial pressure (P_{O_2}) is 159.1 mmHg (20.93% of 760 mmHg), and carbon dioxide's partial pressure (P_{CO_2}) is 0.2 mmHg (0.03% of 760 mmHg).

In the human body, gases are usually dissolved in fluids, such as blood plasma. According to **Henry's law**, gases dissolve in liquids in proportion to their partial pressures, depending also on their solubilities in the specific fluids and on the temperature. A gas's solubility in blood is a constant, and blood temperature also remains relatively constant at rest. Thus, the most critical factor for gas exchange between the alveoli and the blood is the pressure gradient between the gases in the two areas.

Gas Exchange in the Alveoli

Differences in the partial pressures of the gases in the alveoli and the gases in the blood create a pressure gradient across the respiratory membrane. This forms the basis of gas exchange during pulmonary diffusion. If the pressures on each side of the membrane were equal, the gases would be at equilibrium and would not move. But the pressures are not equal, so gases move according to partial pressure gradients.

Oxygen Exchange

The PO_2 of air outside the body at standard atmospheric pressure is 159 mmHg. But this pressure decreases to about 105 mmHg when air is inhaled and enters the alveoli, where it is moistened and mixes with the air in the alveoli. The alveolar air is saturated with water vapor (which has its own partial pressure) and contains more carbon dioxide than the inspired air. Both the increased water vapor pressure and increased partial pressure of carbon dioxide contribute to the total pressure in the alveoli. Fresh air that ventilates the lungs is constantly mixed with the air in the alveoli while some of the alveolar gases are exhaled to the environment. As a result, alveolar gas concentrations remain relatively stable.

The blood, stripped of much of its oxygen by the metabolic needs of the tissues, typically enters the pulmonary capillaries with a PO_2 of about 40 mmHg (see figure 7.6). This is about 60 to 65 mmHg less than the

Key terms

Icons

Key terms are in bold in the text, listed at the end of each chapter, and defined in the glossary at the end of the book. At each chapter's end, you will also find study questions to test your knowledge of the chapter's contents and a reminder of the study guide activities that are available, along with the web address of the online study guide.

At the end of the book is a comprehensive glossary that includes definitions of all key terms, a listing of numbered references for the sources cited in each chapter, and a thorough index. Finally, printed on the inside front and back covers for easy reference are lists of common abbreviations and conversions.

Instructors using this text in their courses will find a wealth of updated ancillary materials available at

www.HumanKinetics.com/PhysiologyOf SportAndExercise, including an instructor guide, a presentation package, a test package, chapter quizzes, and an image bank. The instructor ancillaries also include convenient access to the animations, video clips, and audio clips.

You might read this book only because it is a required text for a course. But we hope that the information will entice you to continue to study this relatively new and exciting area. We hope at the very least to further your interest in and understanding of your body's marvelous abilities to perform various types and intensities of exercise and sports, to adapt to stressful situations, and to improve its physiological capacities. This book is useful not only for anyone who pursues a career in exercise or sport science but also for anyone who wants to be active, healthy, and fit.

This is the case even when elderly subjects are exposed to what we would consider a mild cold stress.⁷ However, people can easily offset these decrements by wearing clothing that is appropriate for the environmental conditions and activity level. By performing so-called behavioral thermoregulation, older athletes can offset the decrements in physiological thermoregulation and continue to exercise safely in cold environments.

During exposure to high altitude, there is little reason to expect older athletes to respond differently than their younger counterparts. Unfortunately, data are lacking regarding aging and the rate and magnitude of acclimatization to altitude. Likewise, it is unclear whether aging per se increases the incidence of any of the altitude illnesses. We can expect the performance of an older athlete at altitude to be similar to that of a younger athlete of comparable physical fitness.

Longevity

Regular physical activity is an important contributor to good health. Does training throughout adulthood affect longevity? Because the aging rate in rats is more rapid than in humans, they have been used as subjects in studies conducted to determine the influence of chronic exercise (training) on **longevity** (the length of one's life). A study by Goodrick¹¹ demonstrated that rats that exercised freely lived about 15% longer than sedentary rats. But an investigation at Washington University in St. Louis showed no significant increase in the life span of rats that voluntarily ran on an exercise wheel.¹² More of the active rats lived to old age, but on average, they still died at the same age as their sedentary counterparts. It is interesting that rats that had restricted food intake and maintained a lower body weight lived 10% longer than the freely eating, sedentary rats. Although exercise training is a key component to energy balance, the only known way to increase longevity is through caloric restriction.

Of course, we cannot directly apply these findings to humans, but these results raise some interesting ques-

tions that might be relevant to our health and longevity. Although it is true that an endurance exercise program can reduce a number of the risk factors associated with cardiovascular disease, only limited information supports the contention that people will live longer if they exercise regularly. Data collected from the alumni at Harvard University and the University of Pennsylvania and from participants at the Aerobic Center in Dallas suggest that there is a decrease in mortality rate and a small increase in longevity (about 2 years) among people who remain physically active throughout life. At a minimum, regular physical activity can increase the **health span**—the number of years of generally healthy living, free from serious disease or chronic disability.

In Review

- The impaired ability of older individuals to tolerate exercise in the heat is due to reduced $\dot{V}O_{2\max}$ and impaired cardiovascular adaptations rather than a direct effect of aging on thermoregulatory control or sweating.
- Regular exercise training can increase skin blood flow and sweating rate and improve the redistribution of cardiac output in older individuals as well as young men and women.
- Older people generally have an impaired ability to tolerate cold, but they can compensate by wearing appropriate clothing.
- Adaptation to altitude appears to be independent of age.
- An active lifestyle appears to be associated with a small increase in longevity. Just as important, an active lifestyle leads to a higher quality of life!
- There is an increased risk of injury from exercise as people age, and injuries tend to be slower to heal.
- The risk of death during exercise is not increased in those who are regularly active but is increased in those who seldom exercise.

In Review

IN CLOSING

In this chapter, we examined the effects of aging on physical performance. We evaluated changes in cardiorespiratory endurance and strength with age. We considered the effect of aging on body composition, which we know can affect performance. And yet, in the course of our discussion, it became clear that much of the change that occurs with aging is attributable to the inactivity that often accompanies aging. When older people participate in training, most of the changes associated with aging are lessened, and the resulting degree of change is similar to that seen in young and middle-aged adults. Thus, we have dispelled many of the myths about the capacity for physical activity of older people.

In the next chapter, we turn our attention to women, who as a group are often considered less capable of physical activity than men. We consider the physiology of girls and women, the impact of this physiology on athletic ability, how performances of female athletes compare with those of male athletes, and special issues associated with being female.

KEY TERMS

cardiovascular deconditioning	maximal expiratory ventilation ($\text{VO}_{2\text{max}}$)
endothelial dysfunction	osteopenia
forced expiratory volume in 1 s ($\text{FEV}_{1\text{s}}$)	osteoporosis
health span	peripheral blood flow
longevity	primary aging

STUDY QUESTIONS

1. What changes occur in height, weight, and body composition with aging? What accounts for these changes? How do these changes affect maximal oxygen uptake?
2. What changes occur in muscle with aging? How do they affect strength and athletic performance?
3. Describe the changes in HR_{max} with age. How does training alter this relationship?
4. How does aging affect maximal stroke volume and maximal cardiac output? What mechanisms can potentially explain these changes?
5. How does the respiratory system change with aging? What happens to vital capacity, $\text{FEV}_{1\text{s}}$, or residual volume, total lung capacity, and the ratio RV/TLC?
6. $\text{VO}_{2\text{max}}$ declines with age across the entire population. Describe the physiological mechanisms that account for this decline. How do trained older individuals maintain a relatively high $\text{VO}_{2\text{max}}$?
7. How do aging and habitual exercise affect blood vessel function?
8. How does age affect anaerobic function?
9. Differentiate between biological aging and physical inactivity.
10. What influence do aging and training have on body composition?
11. Describe the trainability of the older individual for both strength and aerobic endurance.
12. What can be done to minimize age-related losses in motor function and mobility?
13. Describe the changes in strength and endurance performance records with aging.
14. What concerns should we have about older people exercising in hot and cold environments or at altitude?
15. What is the difference between life span and health span?

STUDY GUIDE ACTIVITIES

In addition to the activities listed in the chapter opening outline, two other activities are available in the web study guide, located at

www.HumanKinetics.com/PhysiologyOfSportAndExercise

The **KEY TERMS** activity reviews important terms, and the end-of-chapter **QUIZ** tests your understanding of the material covered in the chapter.

Student and Instructor Resources

Student Resources

Students, visit the free web study guide at www.HumanKinetics.com/PhysiologyOfSportAndExercise for interactive learning activities—all of which can be conducted outside the lab or classroom—as well as animations, video clips, and audio clips to aid your learning. You'll be able to apply key concepts by conducting experiments and recording your own physiological responses to exercise. The guide includes activities and quizzes that test your knowledge of the material as you prepare for classroom quizzes or tests. You'll also have access to links to professional journals and information on organizations and careers in the field.

Updated for the seventh edition, the web study guide includes the following multimedia content:

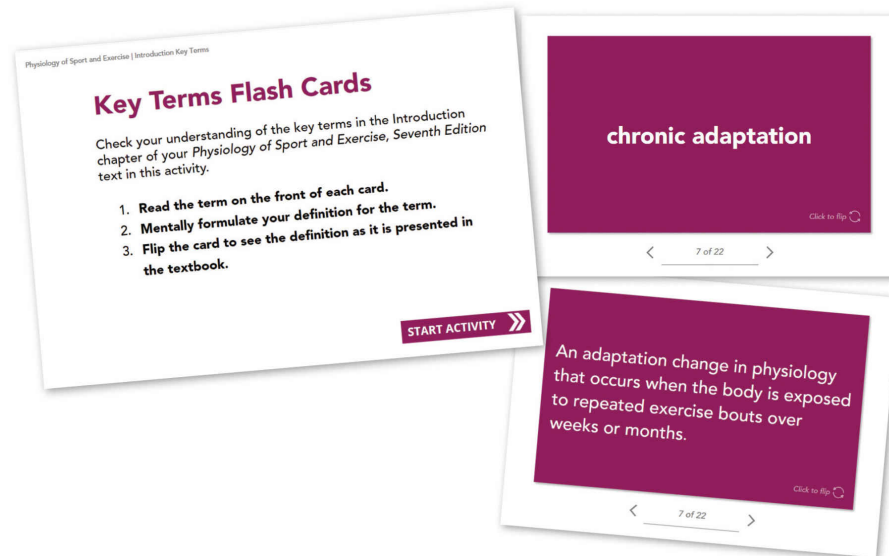
-  26 animated versions of artwork from the text that will help you to understand physiological processes
-  27 video discussions with experts in the field of exercise physiology
-  66 audio clips that describe the processes shown in figures

Look for the icons in the text to know when this additional content is available. As you work to understand a concept illustrated in a figure, refer to the audio or animation for an explanation and to build your understanding. In combination with the web study guide activities, the animations, video, and audio allow you to practice, review, and develop knowledge and skills about the physiology of sport and exercise.

Instructor Resources

Instructor Guide

Specifically developed for instructors who have adopted *Physiology of Sport and Exercise, Seventh Edition*, the instructor guide includes sample lecture outlines, key points, and student assignments for every chapter in the text, along with sample laboratory exercises and direct links to a range of detailed sources on the internet.



Test Package

The test package includes a bank of 1,609 questions created especially for *Physiology of Sport and Exercise, Seventh Edition*. Various types of questions are included: true or false, fill in the blank, essay, and multiple choice. The test package is available for use through multiple formats, including a learning management system, Respondus, and rich text.

Chapter Quizzes

Updated for the seventh edition, these ready-to-use quizzes test students' understanding of the most important concepts in each chapter. Chapter quizzes can be imported into learning management systems or printed for use as written quizzes.

Presentation Package

The presentation package includes a comprehensive series of PowerPoint slides for each chapter. Slides of learning objectives present the major topics covered in each chapter, text slides list key

points, and illustration and photo slides contain graphics found in the text. The presentation package has more than 1,000 slides that can be used directly with PowerPoint and for printing transparencies or slides or making copies for distribution to students. Instructors can easily add to, modify, or rearrange the order of the slides as well as search for slides based on key words. You may access the presentation package by visiting www.HumanKinetics.com/PhysiologyOfSportAndExercise.

Image Bank

The image bank includes most of the illustrations, artwork, and tables from the text, sorted by chapter. These are provided as separate files for easy insertion into tests, quizzes, handouts, and other course materials, which provides instructors with greater flexibility when creating customized resources.

Instructor Video, Animations, and Audio

Within the instructor resources, the multimedia content in the web study guide is compiled for convenient access and inclusion in lectures and classroom presentations.

The instructor guide, test package, chapter quizzes, presentation package, image bank, video clips, animations, and audio clips are free to course adopters.

Acknowledgments

We would like to thank the staff at Human Kinetics for their continued support of the seventh edition of *Physiology of Sport and Exercise* and their dedication to publishing a high-quality product that meets the needs of instructors and students alike. Recognition goes to Amy Tocco (acquisitions editor) as well as our capable developmental editors: Lori Garrett (first edition), Julie Rhoda (second and third editions), Maggie Schwarzentraub (fourth edition), and Kate Maurer (fifth and sixth editions); Judy Park took over the reins as developmental editor for this seventh edition and has worked tirelessly and expertly to keep all phases of the project on schedule while continuing to demand the high quality for which our book is known. They have all been a true pleasure to work with, and their competence and skill are evident throughout the book. Special thanks go to Joanne Brummett for her artistic expertise and contributions to continuously improving the artwork.

For the seventh edition, special thanks also go to a handful of colleagues who provided their valued expertise and time. In particular, direct feedback and input from Drs. Gustavo Nader, Jinger Gottschall, Lacy Alexander, and Jim Pawelczyk at Penn State were invaluable in making substantive changes that not only updated and enhance the content but also provided high quality feedback from an instructor's viewpoint. Special recognition goes to the "postdoc dream team" of Drs. Jody Greaney and Anna Stanhewicz for all of their hard work in helping update all of the Research Perspective elements. In addition to Larry Kenney's Penn State colleagues, thanks also go to Dr. Bob Murray, who once again contributed his vast knowledge about ergogenic aids to [chapter 16](#).

Finally, we thank our families for their constant love, support, and patience while we were writing, rewriting, editing, and proofing this book across all seven editions.

W. Larry Kenney
David L. Costill
Jack H. Wilmore (posthumously)

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